

Elastic Mass Model Approach for Fuel Dynamics in Microgravity Environments

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<https://doi.org/10.2514/1.A36218>

Managing fuel in the challenging conditions of zero gravity necessitates a profound grasp of fluid dynamics, particularly the pivotal role of surface tension. This study introduces a novel mechanical model to replicate fuel dynamics within a partially filled tank in zero gravity. Unlike traditional models, this model, with its six degrees of freedom and the implication of realistic liquid parameters, effectively captures the stochastic behavior of propellant mass. The model accounts for surface tension effects and elastic collisions with tank walls during orbital maneuvers. This novel elastic mass model overcomes the limitations of existing methods like the constrained surface model, pendulum-based, and spring-mass models, by leveraging a conceptual analogy rooted in Hooke's Law and improved Hertz theory. Established simulations have demonstrated the model's capability to imitate wobbling fuel motions and the resulting forces on orbiting spacecraft. Comparison with computational fluid dynamics results showed the auspicious potential of the model.

Nomenclature

a	=	contact width between the fuel mass and the tank wall
B	=	fuel model bulk modulus
C_k, C_d	=	Taylor analogy constants derived from experimental data
C_r	=	collision restitution coefficient
$C_{TA}, K_{TA}, \omega_{nTA}$	=	Taylor analogy damping coefficient, spring constant, and natural frequency
E_f, E_f^c	=	fuel model elasticity and complex elasticity moduli
E_T	=	elasticity modulus of the tank
F, O_F	=	fuel model reference frame and its origin O_F , the fuel mass center
F_j^A	=	generalized active forces of body j
$F_j^I, F_{jNL}^I, F_{jM}^I$	=	generalized inertial force of body j and its nonlinear and linear terms, respectively
f_{ad}, f_c, f_d	=	adhesion, control, and disturbance forces
f_{col}, f_n, f_t	=	collision total, normal and tangential forces
I, J	=	inertia matrices of the spacecraft hub and the fuel ball respectively, $\text{kg} \cdot \text{m}^2$
K, χ	=	collision stiffness coefficient and hysteresis damping factor
M_j	=	total mass matrix of body j
m	=	total mass of the sloshing fuel
m_s	=	spacecraft hub mass
$N(i, j, k)$	=	inertial reference frame at Earth center
n_{nor}, n_{tan}	=	unit vectors of the collision normal and tangential forces

P	=	collision point
q_s, q_f, u_s, u_f	=	generalized coordinates and speeds of the spacecraft and the fuel ball
R_f, R_T	=	fuel model and tank radii
R_{312-f}, R_{312-T}	=	transformation matrices from the fuel and tank frames to the inertial frame
$r_{col}, r_{col,T}, r_{col,F}$	=	collision point position vectors in the inertial, tank, and fuel frames
$\dot{r}_{col,F}^N$	=	collision point velocity in the fuel reference expressed in the inertial reference
r_f, r_s, r_T	=	position vectors of the fuel, spacecraft, and tank centers in the inertial frame
T, O	=	spacecraft and tank reference frame and its origin O the tank center
v_j, ω_j	=	overall linear and angular velocities of body j
v_{j-i}, ω_{j-i}	=	partial linear and angular velocities of body j
v_s	=	threshold velocity for the dynamic friction
v_t	=	velocity tangential component
x_j, y_j, z_j	=	Cartesian coordinates in inertial frame of body j
γ	=	effective surface energy per unit area
$\delta, \dot{\delta}, \dot{\delta}^{(-)}$	=	normal embedding depth vector and the norms of its velocity and initial velocity
η	=	collision force coefficient generally equal to 2/3
μ	=	fuel viscosity
μ_v, μ_d, μ_s	=	coefficients of the static, dynamic, and viscous frictions
ν_f, ν_T	=	Poisson ratio of fuel model and the tank
ρ, σ	=	density and surface tension of the sloshing fuel
$\varphi_j, \theta_j, \psi_j$	=	Euler angles of the spacecraft and the fuel ball

Subscripts

f	=	fuel index
s	=	spacecraft index
T	=	tank index

I. Introduction

L IQUID sloshing dynamics is a multidisciplinary field of research with significant applications not only in aerospace engineering but also in civil engineering and seismology. The behavior of liquids, particularly under the unique conditions of microgravity, also known as the state of weightlessness, presents

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complex challenges that are critical to the stability and maneuverability of spacecraft and satellites [1]. This research contributes to the understanding of these challenges by focusing on the dynamics of liquid in zero gravity, where conventional models fall short. Previous studies have extensively explored the deterministic aspects of slosh dynamics. However, less attention has been paid to stochastic behaviors, especially under conditions where the excitation frequencies exceed certain thresholds, leading to random instabilities and potentially catastrophic outcomes [2]. Under microgravity, the absence of hydrostatic pressure and gravitational forces introduces additional complexities [3]. Liquids behave differently, exhibiting phenomena such as containerless floating and significant shifts in the center of mass [4]. These behaviors pose severe risks to precise spacecraft control and navigation. For the purposes of incorporating the dynamic effects of sloshing in the spacecraft control and stability analysis, it is convenient to replace the liquid conceptually by an equivalent linear mechanical system. This approach simplifies the analysis by substituting the complex fluid dynamics equations with the more manageable equations of motion for oscillating point masses and rigid bodies. Such a substitution is crucial because coupling the motion equations of a flexible spacecraft with those of a continuous liquid remains computationally prohibitive, even with the advancements in supercomputing, making it unsuitable for standard design analyses [5–8]. Research in this field has traditionally relied on equivalent mechanical models generally representing the portion of fluid moving in unison with the tank as a fixed rigid mass at a specified depth, and the part of the fluid experiencing convective pressure and sloshing at the free surface was approximated in a linear regime using either a set of mass–spring–dashpot systems or a set of pendulums [9]. Certain pendulum models were enhanced to replicate nonlinear fluid oscillation by incorporating cubic springs, which offer enough elasticity to counteract the container’s curvature effects. Additionally, to account for damping in scenarios involving more viscous fluids, a dashpot was also incorporated [10]. However, these models, in addition to the limitations imposed by their restricted degrees of freedom, often fail to adequately capture the nonlinear responses and the critical effects of surface tension, which become dominant in the absence of gravity. Traditional models, such as spring-mass and pendulum models, rely on restrictive assumptions, including fixed pendulum lengths or small-amplitude linearized dynamics. These models often approximate sloshing dynamics with only three degrees of freedom (DOF) and assume small amplitude motions, limiting their applicability to real-world scenarios where sloshing is more complex. Similarly, the Constrained Surface Model and Moving Pulsating Ball Model impose rigid constraints that fail to capture the behavior of fuel in partially filled tanks under microgravity, where the fuel does not always remain in contact with the tank walls as in the cases of free-floating masses, surface tension-driven dynamics, and noncontact wobbling. The pulsating behavior is typically assumed to be isotropic (uniform in all directions), which overlooks the anisotropic (direction-dependent) nature of sloshing under strong directional excitation [11,12]. Indeed, surface tension causes the liquid to behave like an elastic body, leading to unexpected reorientations and distributions of the liquid mass within the tank [13]. Parameters and constants expressions of the aforementioned models have been determined in numerous studies, for numerous shapes of containers such as rectangular, cylindrical, spherical, and ellipsoidal [14]. However, for zero gravity environments where the liquid motion is inertia dominated, most of studies relied on numerical analysis for accurate predictions, as demonstrated by Navickas and Cheng [15] or few experimental results such as those done in the Spheres project of NASA [16]. Nevertheless, for small tank motions, such as those occurring during spacecraft station-keeping, understanding the sloshing frequencies and dynamics in low-gravity conditions remains essential. Dodge [5] and Carney [17], among the very few researchers who analytically investigated the zero gravity liquid dynamics, determined the parameters of mechanical models for ellipsoidal and rectangular tanks, however, they did not consider cylindrical and spherical tanks for being exceptions]. In fact, in zero gravity, the liquid interface

forms either a sphere or a sector of a sphere, depending on the contact angle, a complete sphere if the contact angle is 0 deg. The gas ullage bubble might need to be *swallowed* within the tank to fit, or it may form an annular interface that intersects both the top and bottom of the tank. However, from geometrical considerations and regardless of fill level, this phenomenon is different for cylindrical and spherical tanks [5].

In addition to theory, the fascinating feature of liquids behaving as wobbling masses in zero gravity have been witnessed in reality in many occasions. During the Commercial Resupply Services (CRS 5) mission, records showed clouds of liquid oxygen bubbles formed in the Falcon 9’s first- and second-stage tanks after separation and continued moving for at least 6 min, coalescing with each other [18]. Also, scientists on the International and Tiangong space stations, used hydrophobic paddles, to prevent liquid adhesion during dynamic processes, allowing them to pass a sphere of water back and forth, demonstrating the inertia-dominated motions. They showed that the larger the drop, the less force is needed to break it up. Conversely, the smaller the drop, the more force it can withstand. These wobbling masses are significantly larger than drops in normal gravity, being over 100 times the size of a raindrop [19,20].

This paper introduces a novel simplified mechanical model with 6 deg of freedom, designed to address the dynamic challenges posed by liquid propellant in a spacecraft’s fuel tank under zero gravity conditions. Notably, in such an environment, the tendency of liquids to form spherical shapes due to surface tension means they lend themselves to analogous treatments with solid elastic materials. This analogy allows us to employ established mechanical models and equations from solid mechanics to predict the interactions between the propellant and the tank walls during maneuvers. By leveraging the Young–Laplace equations for surface tension, Hooke’s Law for elastic behavior, and the improved Hertz theory for collision forces computation, our model offers a comprehensive framework for understanding and predicting fuel slosh dynamics in space in zero gravity. Sections II and III, delineate the rationale and methodology of the liquid–elastic body analogy, which provides a framework for calculating the collision forces between the tank wall and the fuel mass. The enhanced Hertz theory, known as the Flores contact model, applied in Sec. IV, is a novel method for describing the force for both soft and hard contacts. This approach is chosen for being easy to implement in any multibody code for dynamic analysis and accounts for the material, mechanical, geometrical, and kinematical properties of the contacting bodies [21]. Theories like the Johnson–Kendall–Roberts (JKR) and Derjaguin–Muller–Toporov models further aid in examining the adhesion between the propellant mass and the tank walls [22,23]. JKR is briefly introduced in Sec. IV. In Sec. V, the equations of motion for a partially fuel-filled rigid spacecraft is derived by Kane’s method, which simplifies computational complexity and integrates well with numerical methods and is a highly efficient approach for deriving the equations of motion in complex mechanical systems with multiple degrees of freedom and nonholonomic constraints [24]. Different simulations results are interpreted and discussed in detail in the Sec. VI, demonstrating the model’s efficacy in capturing the observed complex dynamics of fuel sloshing under zero gravity, offering promising avenues for improving spacecraft design and control systems in such conditions. In Sec. VII, we concluded with a few remarks by qualitatively and quantitatively observing the results of a number of simulations. The elastic fuel ball behavior has been observed in both hydrophobic and adhesive tanks. And the effect of the rigid–flexible coupled dynamics on an orbiting spacecraft attitude is presented. Certain results are compared with computational fluid dynamics (CFD) results showing strong compliance.

The key contributions of our elastic mass model are as follows:

- 1) Unlike all other mechanical models, this model has full six degrees of freedom, allowing it to capture all translational and rotational motions of the fuel.
- 2) Thanks to overcoming the traditional geometrical restrictions, the elastic mass model is able to mimic the in-space witnessed wobbling behaviour exhibiting clearly the surface tension dominance in zero gravity.

3) This simplified mechanical model harnesses a liquid–elastic body analogy using surface tension dominance by employing Hooke’s law and the elasticity theory with ensuring to fully encapsulate the real properties of the fuel when converted into mechanical parameters by employing the Taylor Breakup Model.

4) Flores Model, the enhanced Hertz contact theory, is applied, which incorporates a damping term to simulate fuel bulk collision during the impact events.

5) Comprehensive validation of the novel model with simulations that replicate real-life space phenomena is conducted, demonstrating strong compliance with CFD results and experimental observations.

II. Theoretical Framework and Assumptions

A. Surface Tension Fundamental Notion

Generally, gases and liquids are defined by the fact that they take the shape of their container. However, for a better understanding and for more accuracy, we emphasize in this paper that the container’s walls are in fact the ones who limit the fluid workspace and expansion rather than the fluid itself seeking to take the shape of the container. Actually, liquids tend to maintain equilibrium with the minimum amount of energy in any container of any shape.

Their free surface tries to take the *laziest form*, a form that does not require much of internal work. This work is mainly presented in breaking the least numbers of the enthalpy and entropy driven intermolecular bonds (do not confound with chemical bonds), such as the electrostatic attractions (Coulomb forces) and van der Waals/London dispersion forces. In other terms, as shows Fig 1, liquids molecules at the free surface level have fewer bonds to make. And to make a larger free surface area, the liquid needs to break more bonds of internal molecules so they get arranged in the surface. The needed energy to do that is known by surface tension. This explains, in the absence of external forces, the spherical shape of liquids surface because it has the smallest surface and laziest form possible. The aforementioned bindings are also known by the term capillary forces. It is called cohesion if the binding occurs between molecules of the same matter and adhesion if between unlike molecules such as those seen between a liquid and its container’s wall.

B. Mechanical Modelling Equivalence in Dynamic and Static States

Mechanical models are considered equivalent if they exhibit to a maximal extent the behavior of the modeled system in both static and dynamic states. In brief, the following principles should be followed in establishing mechanical equivalent model; that is, the EMM (Equivalent Mechanical Model) should have approximately the same natural frequencies, same mass quantities, same mass center (COM) positions in static state, and same forces and torques in the dynamic state as the real system.

Natural frequency is one of the most important intrinsic properties of liquid sloshing. When a liquid completely fills a capped container, it behaves like a rigid body under dynamic conditions [25]. However, when the liquid surface is free to move, the resultant sloshing introduces natural frequencies, necessitating a mechanical model with oscillating masses rather than rigid body as they can easily incorporate damping [26] and are independent of tank shape or fill level [5]. In most literature, the pendulum model is preferred

because its natural frequency adjusts with the variation of the axial acceleration such as gravity, unlike the spring-mass model which requires adjusting the spring constant accordingly. However, in zero gravity, the sloshing natural frequency is nonzero but generally lower than what it is in normal gravity due to the smaller surface tension force compared to gravitational or thrust-induced forces which leads to large liquid inertia-dominated motions even when excited slightly, making the high-gravity linear motion assumptions less valid [5]. Masses quantities and COM positions are considered on an equal basis as the natural frequency in establishing mechanical equivalent modes. Classically, the liquid is modeled by a rigidly attached mass in addition to one or more oscillating mass. The total quantity of masses should be equal to the total mass of the liquid during all regimes, and in the static state, the mass centers bary-center should coincide with the position of the mass center of the total liquid. However, the choice of the repartition of the mass quantities and the mass centers locations depends on the model. In addition, the equivalent mechanical model should exhibit to a maximal extent the same forces of the real liquid such as the capillary, viscous, pressure, and inertial forces including the centrifugal ones and the gravitational if they exist.

Finally, the boundary conditions and some assumptions must also be considered in equivalent model. As it is, the liquid cannot penetrate the tank wall, and the tank wall is nondeformable; only the elastic fuel model is deformable, and the fuel ball breakup occurs when the deformation is equal to the fuel ball original radius. The fuel tank is rigidly attached to the spacecraft and no backlash is considered. In normal circumstances, the liquid is assumed to be an incompressible Newtonian liquid, no convection or thermal gradients are to be considered. Some coefficients values are reasonably assumed such as the collision restitution coefficient; the viscosity; dynamic and static friction coefficients μ_v , μ_d , and μ_s ; and the threshold velocity for the dynamic friction v_s (these entities are expressed in Sec. 4).

III. Liquid Propellant-Elastic Body Analogy

For our purposes, the liquid propellant as shown in Fig. 2a is conceptualized as a spherical elastic body in Figs. 2b and 2a simplification inspired by the observed dominance of surface tension. However, an immediate challenge in the application of this model is the accurate translation of the liquid’s physical properties into those applicable to an elastic body, such as Young’s modulus.

Direct relationships linking a liquid’s surface tension to Young’s modulus are not readily available; however, there exists a correlation between surface tension and the bulk modulus of a fluid. This relationship, while informative, does not fully encapsulate the real properties of the fuel when converted into parameters for an elastic body model. The absence of critical data, like the shear modulus, further complicates this translation.

To bridge the gap between theoretical correlations and practical application, the Taylor Breakup Analogy (TAB) model offers a valuable intermediary as shown in Fig. 3. From the spring mass system, the stiffness coefficient, a parameter easily derived from observing the sloshing frequencies and amplitudes, is used to estimate the effective Young modulus of the liquid modeled as an elastic sphere. The formula linking spring stiffness to Young’s modulus, typically used for solid materials, is adapted here to approximate the propellant’s elastic properties.

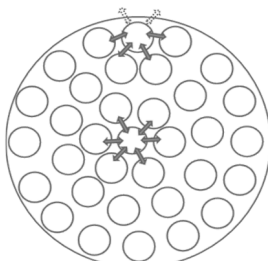
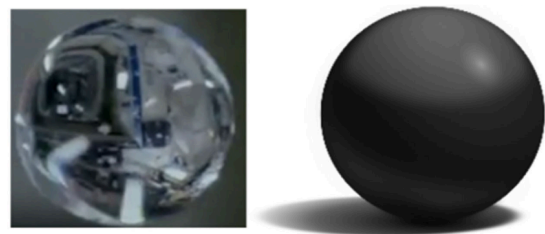


Fig. 1 Molecular bonds, origin of the spherical shape of liquids in zero gravity.



a) Liquid ball b) Elastic spherical model

Fig. 2 Liquid–elastic body analogy in zero gravity.

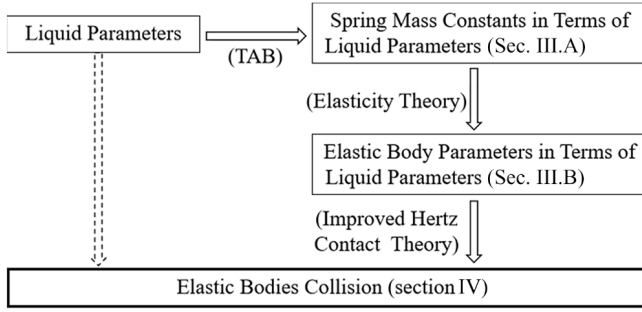


Fig. 3 Methodology diagram.

A. Liquid Propellant–Spring Mass Analogy

This analogy is particularly relevant under conditions where surface tension predominates over other external forces, such as in microgravity environments typical in space. It provides a physical basis for understanding how surface tension influences the stability and behavior of an oscillating liquid bulk in absence of gravity and simplifies the complex fluid dynamics into a more manageable spring-mass system, drawing parallels between their oscillatory properties, incorporating as many of the liquid's real parameters as possible, thus minimizing the assumptions required in the spherical ball model.

From Taylor's analogy, as shown in Fig. 4, the spring constant K_{TA} and the damping coefficient C_{TA} are expressed as follows:

$$K_{TA} = \frac{C_k m \sigma}{\rho R_f^3} \quad (1)$$

$$C_{TA} = \frac{C_d m \mu}{\rho R_f^2} \quad (2)$$

$$\omega_{nTA} = \sqrt{\frac{K_{TA}}{m}} \quad (3)$$

Here, C_k is a constant derived from experimental data, m is the mass of the oscillating liquid, σ is the surface tension of the liquid, ρ is the density, μ is the viscosity, R_f is the radius of the fuel mass which can be calculated from the fuel volume and density, C_d represents a damping constant which also originates from experimental observations, and ω_{nTA} is the corresponding natural frequency.

B. Spring Mass–Elastic Body Analogy

Using the derived parameters from TAB, the Young modulus E_f of the liquid modeled as an elastic sphere can be estimated based on classical definitions from elasticity theory

$$E_f = \frac{\text{stress}}{\text{strain}} = \frac{\text{force/section area}}{\text{change in length/initial length}} \quad (4)$$

The spring stiffness K_{TA} represents the relation between the force and the length deformation, and the Young modulus is then written as

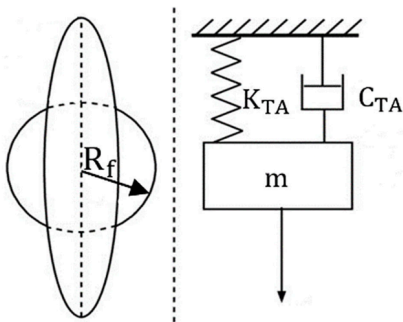


Fig. 4 Schematic representation of the Taylor Analogy breakup.

$$E_f = K_{TA} \frac{\text{initial length}}{\text{section area}} \quad (5)$$

For a spherical model, the geometric factor can be addressed as a gap in spherical Young's modulus calculations and might be more complex than for a simple rod. Therefore, an approach is assumed; the initial length can be considered as a characteristic dimension, such as the diameter, and the section area could be the cross-sectional area at a typical slice of the sphere such as the equator or the average cross-sectional area along the diameter. Each slice's area is given by πr^2 with r being the radius of the cross-sections the long of the diameter. Finally, the Young modulus of the fuel ball is written as

$$E_f = K_{TA} \frac{3}{\pi R_f} \quad (6)$$

However, the calculation of the Young's modulus of a spring mass system can be assumed depending on the damping significance. If damping is not assumed as significant (i.e., the fuel ball behaves more elastically with minimal energy loss), E_f effectively becomes equivalent to the Young's modulus as in Eq. (6). In case the damping is significant, C_{TA} also impacts the material properties, and using a complex modulus E_f^c might be necessary to account for energy loss characteristics. In viscoelastic theory, the complex modulus is given as a combination of an elastic component and a damping component representing the storage and loss moduli as

$$E_f^c = K_{TA} \frac{3}{\pi R_f} + i \omega_{nTA} C_{TA} \quad (7)$$

$$|E_f^c| = \sqrt{\left(K_{TA} \frac{3}{\pi R_f}\right)^2 + (\omega_{nTA} C_{TA})^2} \quad (8)$$

where i represents here the imaginary unit and the spring constant K_{TA} is representing the elastic properties (storage modulus). The damping coefficient C_{TA} is representing the viscous properties (loss modulus). And the combination $\omega_{nTA} C_{TA}$ indicates how the viscous damping contributes to the system's response as a function of the frequency of oscillation. The magnitude $|E_f^c|$ provides a comprehensive measure of how stiff and how resistive to oscillations the material system is. It accounts for both the material's ability to store energy (via the springlike behavior) and dissipate energy (via the dashpotlike behavior).

Poisson's ratio ϑ_f is another relevant fundamental material property for the fuel elastic model. Generally, a perfectly incompressible isotropic material deformed elastically at small strains would have a Poisson's ratio of exactly 0.5. It is defined following Lamé's relation in terms of the Young's and the fuel's bulk moduli B as

$$\vartheta_f = \frac{1}{2} \left(1 - \frac{E_f}{3B}\right) \quad (9)$$

IV. Body Collision and Flores Contact Force Model

This section discusses the mathematical modeling of collision dynamics between the fuel ball and the tank wall using the Flores contact force model, an enhancement of the Hertzian contact theory that incorporates energy dissipation [27]. Vectors are written in bold italic format, while scalars' and vectors' magnitudes are written in regular italic font.

The collision force $\mathbf{f}_{col}$ between the fuel ball and the tank wall is calculated in an only contact point P , and its position vectors $\mathbf{r}_{col}$, $\mathbf{r}_{col,T}$, and $\mathbf{r}_{col,F}$ in the inertial reference frame N , tank reference frame T , and fuel reference frame F , respectively, are

$$\mathbf{r}_{col} = \mathbf{r}_f + (R_f - \delta) \mathbf{n}_{nor} \quad (10)$$

$$\mathbf{r}_{col,T} = \mathbf{R}_{312-T}(\mathbf{r}_{col} - \mathbf{r}_T) \quad (11)$$

$$\mathbf{r}_{\text{col},F} = \mathbf{R}_{312-f}(\mathbf{r}_{\text{col}} - \mathbf{r}_f) \quad (12)$$

where $\mathbf{r}_f$ and $\mathbf{r}_T$ are the position vectors of the fuel and tank centers in the inertial frame and $\mathbf{R}_{312-f}$ and $\mathbf{R}_{312-T}$ are, respectively, the Euler angles transformation matrices from the fuel ball and fuel tank to the inertial frame,

$$\mathbf{R}_{312-i} = \begin{bmatrix} c(\theta_i)c(\psi_i) - s(\varphi_i)s(\theta_i)s(\psi_i) & -s(\psi_i)c(\varphi_i) & s(\theta_i)c(\psi_i) + s(\varphi_i)s(\psi_i)c(\theta_i) \\ c(\theta_i)s(\psi_i) + s(\varphi_i)s(\theta_i)c(\psi_i) & c(\varphi_i)c(\psi_i) & s(\theta_i)s(\psi_i) - s(\varphi_i)c(\theta_i)c(\psi_i) \\ -s(\theta_i)c(\varphi_i) & s(\varphi_i) & c(\varphi_i)c(\theta_i) \end{bmatrix} \quad (13)$$

where φ_i , θ_i and ψ_i are the Euler angles, c and s refer to cosine and sine, the index i here refers to the fuel and the spacecraft, δ is the normal embedding depth caused by the collision and is defined by

$$\delta = \|\mathbf{r}_T - \mathbf{r}_f\| - (R_T + R_f) \quad (14)$$

and $\mathbf{n}_{\text{nor}}$ is the unit vector oriented from the tank's center O to the contact point P as the direction of the normal to the collision surface, which, in the case of a sphere inside another sphere, relates the two spheres centers and the collision point as well; its unit vector is

$$\mathbf{n}_{\text{nor}} = \frac{(\mathbf{r}_T - \mathbf{r}_f)}{\|\mathbf{r}_T - \mathbf{r}_f\|} \quad (15)$$

The collision force $\mathbf{f}_{\text{col}}$ is written as the sum of a normal and tangential components $\mathbf{f}_n$ and $\mathbf{f}_t$, respectively,

$$\mathbf{f}_{\text{col}} = \mathbf{f}_n + \mathbf{f}_t \quad (16)$$

We are now using Flores contact model, which is an improvement of the Hertzian impact theory, so it is applicable for systems with energy loss as shown in Fig. 5. In this improved model, the term χ is added, and it represents a hysteresis damping factor [21]. The collision normal force $\mathbf{f}_n$ is given as

$$\mathbf{f}_n = K\delta^\eta \left(1 + \frac{8(1-C_r)}{5C_r\dot{\delta}^{(-)}} \dot{\delta} \right) \mathbf{n}_{\text{nor}} \quad (17)$$

where K , the collision stiffness coefficient, is given as

$$K = \frac{43}{3} \left(\frac{1 - \vartheta_T^2}{E_T} + \frac{1 - \vartheta_f^2}{E_f} \right) \sqrt{\frac{R_T R_f}{R_T + R_f}} \quad (18)$$

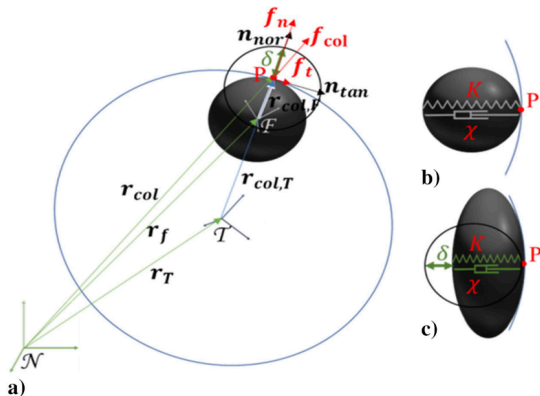


Fig. 5 Schematic representation of Flores contact model: a) collision scheme, b) nondeformed elastic model, and c) deformed elastic model.

and the damping factor is defined as

$$\chi = \frac{8(1-C_r)K}{5C_r\dot{\delta}^{(-)}} \quad (19)$$

Where η is the force coefficient generally equal to 2/3; C_r is the restitution coefficient assumed to be equal to 0.7, which is relatively

large because of high elasticity; and $\dot{\delta}$ and $\dot{\delta}^{(-)}$ are the magnitudes of the normal embedding depth velocity and initial velocity, respectively. The normal embedding depth velocity vector $\dot{\delta}$ is the projection on the normal direction of the velocity of the contact point P in the fuel reference expressed in the inertial reference as $\dot{\mathbf{r}}_{\text{col},F}^N$,

$$\dot{\delta} = \dot{\mathbf{r}}_{\text{col},F}^N \mathbf{n}_{\text{nor}} \mathbf{n}_{\text{nor}} \quad (20)$$

Where ϑ_f , ϑ_T , E_f , and E_T are the material properties of the fuel as introduced in Sec. 3.B, and those of the tank R_f and R_T are their radii.

The tangential force component depends on whether the contact surfaces of the fuel and the tank wall slip or stick. The transition between sticking and slipping is not instantaneous or binary such as in the straightforward model of Coulomb friction. A more nuanced model is often required based on how long and how fast the surfaces have been in contact or moving relative to each other [28].

Once slipping starts, the friction force is defined by the dynamic friction coefficient μ_d . The maximum static friction force that can occur before slipping is given by

$$\mathbf{f}_t = \mu_s \mathbf{f}_n \mathbf{n}_{\text{tan}} \quad (21)$$

where μ_s is the coefficient of the static frictions and $\mathbf{n}_{\text{tan}}$ is the unit vector of the tangential force,

$$\mathbf{n}_{\text{tan}} = -\frac{\Delta \mathbf{v}_t}{\Delta v_t} = -\frac{(\dot{\mathbf{r}}_{\text{col},F} - \dot{\delta})}{\|\dot{\mathbf{r}}_{\text{col},F} - \dot{\delta}\|} \quad (22)$$

with Δv_t being the tangential component of $\dot{\mathbf{r}}_{\text{col},F}$ and the notation Δ is used here because when the two colliding bodies are deformable the difference $\dot{\mathbf{r}}_{\text{col},T} - \dot{\mathbf{r}}_{\text{col},F}$ between both velocities should be calculated. Nevertheless, because the tank is assumed nondeformable, only the fuel deformation velocity is considered.

On the other hand, for sticking conditions, the tangential collision force can be written as

$$\mathbf{f}_t = \left(f_n \left(\mu_d + ((\mu_s - \mu_d) \exp^{-(\Delta v_t/v_s)^2}) + \mu_v \Delta v_t \right) \right) \mathbf{n}_{\text{tan}} \quad (23)$$

where the term $((\mu_s - \mu_d) \exp^{-(\Delta v_t/v_s)^2})$ represents a linear interpolation between dynamic and static friction coefficients with the implication of an exponential decay function to model how the friction force decreases from static to dynamic as the tangential velocity Δv_t increases; v_s is the threshold velocity for the dynamic friction, and the term $\mu_v \Delta v_t$ represents the viscous friction, where μ_v is the viscosity coefficient.

During adhesion, the contact width a following JKR as shown in Fig. 6 can be expressed as

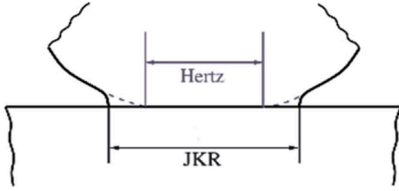


Fig. 6 Adhesion based on JKR model.

$$a^3 = \frac{3R_f[(1 - \theta_T^2/E_T) + (1 - \theta_f^2/E_f)]}{4} \left((f_{\text{col}} + f_{\text{ad}}) + 3\gamma\pi R_f \right. \\ \left. + \sqrt{6\gamma\pi R_f(f_{\text{col}} + f_{\text{ad}}) + (3\gamma\pi R_f)^2} \right) \quad (24)$$

where f_{ad} is the adhesion force and γ is an effective surface energy per unit area. For two dissimilar surfaces, the effective surface energy is given by

$$\gamma = \gamma_f + \gamma_T + \gamma_{Tf} \quad (25)$$

where γ_f , γ_T and γ_{Tf} are respectively the surface energies of the fuel ball, the tank wall, and the interface. This approach is more detailed in Ref. [29].

V. Kane Method for Partially-Fuel-Filled Spacecraft Dynamics in Zero Gravity

As shown in Fig. 7, the spacecraft has a partially-liquid-filled spherical tank, and the floating mass of fuel is modeled by a spherical body. $N(i, j, k)$ denotes the inertial frame. The spacecraft body fixed COM frame T is chosen to coincide with the tank's geometrical center O , and F denotes the fuel ball frame located at its center O_f . Generally, the spacecraft is equipped with flexible systems such as appendages and manipulators; however, in this validation example, the system is restricted to the orbiting rigid hub and the fuel mass. Kane's method is preferred for such complex high DOF mechanical systems for its particularity and effectiveness in handling them and deriving the equations of motion without the need to derive and solve the system's Lagrangian or Hamiltonian, also, for its high integration with numerical methods.

The satellite and the propellant ball are considered as two separate orbiting systems, each with six DOF. Their dynamics are only coupled and interconnected during collision. Therefore, the expressions of the generalized speeds, partial velocities and accelerations, and inertial forces are practically the same for each. The selection of generalized coordinates for the spacecraft q_s and fuel q_f , as well as the reference frames in which they are defined, is not unique. The primary objective is to simplify the system's configuration. Their number is equal to the number of degrees of freedom of the system, thus six for each of the orbiting spacecraft and the floating propellant ball. The generalized coordinates q_s and q_f and speeds u_s and u_f are written then as

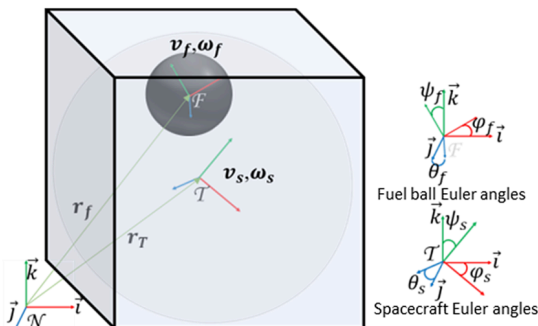


Fig. 7 Schematic representation of partially fuel filled rigid spacecraft and Euler angles.

$$q_s = \begin{bmatrix} q_{s1} & q_{s4} \\ q_{s2} & q_{s5} \\ q_{s3} & q_{s6} \end{bmatrix} = \begin{bmatrix} x_s & \varphi_s \\ y_s & \theta_s \\ z_s & \psi_s \end{bmatrix}; u_s = \dot{q}_s = \begin{bmatrix} u_1 & u_4 \\ u_2 & u_5 \\ u_3 & u_6 \end{bmatrix}; \\ q_f = \begin{bmatrix} q_{f1} & q_{f4} \\ q_{f2} & q_{f5} \\ q_{f3} & q_{f6} \end{bmatrix} = \begin{bmatrix} x_f & \varphi_f \\ y_f & \theta_f \\ z_f & \psi_f \end{bmatrix}; u_f = \dot{q}_f = \begin{bmatrix} u_7 & u_{10} \\ u_8 & u_{11} \\ u_9 & u_{12} \end{bmatrix} \quad (26)$$

The formula to convert the position r_A of a point A, from a rotating frame k (T or F) to the inertial frame N , is written as

$$r_A^N = R_{312-i} r_A^k + r_k^N \quad (27)$$

where the transform matrix R_{312-i} is expressed in Eq. (13) and r_k^N is the position vector of the origin of the frame k .

A. Partial Velocities, Generalized Inertial Forces, and Generalized Active Forces

To break down the motion into manageable parts and systematically formulate the equations of motion, the overall linear and angular velocities v_s , v_f , ω_s , and ω_f of the system are decomposed into partial linear and angular velocities v_{s-i} , v_{f-i-6} , ω_{s-i-3} , and ω_{f-i-9} . Those partial velocities are interpreted as the contributions to the system's velocity due to each isolated generalized speed u_i for $i = 1, \dots, 12$:

$$v_{s-i} = \frac{\partial v_s}{\partial u_i}; i = 1, 2, 3; \omega_{s-i-3} = \frac{\partial \omega_s}{\partial u_i}; i = 4, 5, 6 \quad (28)$$

$$v_{f-i-6} = \frac{\partial v_f}{\partial u_i}; i = 7, 8, 9; \omega_{f-i-9} = \frac{\partial \omega_f}{\partial u_i}; i = 10, 11, 12 \quad (29)$$

The formula to convert the velocity of a point A, from a frame k (T or F) to the inertial frame N is written as

$$v_A^N = v_k^N + v_A^k + \omega^{k/N} \times r_A^k \quad (30)$$

where is $\omega^{k/N}$ the rotation of the frame k in N and r_A^k is the position vector of A in k .

The generalized inertial force F_j^I is divided into two terms, $-F_{jM}^I$ and $-F_{jNL}^I$ [24], where the index j refers to each of the colliding bodies, the spacecraft, and the fuel ball model. F_{jM}^I is the product of total mass matrix M_j , which is the sum of the linear contribution of system's degrees of freedom and the derivatives of the generalized speeds corresponding to those freedoms,

$$F_{sM}^I = M_s \dot{v}_s; F_{fM}^I = M_f \dot{v}_f \quad (31)$$

$$M_s = v_{s-i} m_s v_{s-i} - v_{s-i} m_s r_s \times \omega_{s-i} + \omega_{s-i} m_s r_s \times v_{s-i} \\ + \omega_{s-i} I \omega_{s-i}; i = 1, 2, 3 \quad (32)$$

$$M_f = v_{f-i} m_f v_{f-i} - v_{f-i} m_f r_f \times \omega_{f-i} + \omega_{f-i} m_f r_f \times v_{f-i} \\ + \omega_{f-i} I \omega_{f-i}; i = 1, 2, 3 \quad (33)$$

where I and J are the inertia matrices of the spacecraft and the fuel ball, respectively. The nonlinear terms resulting from those derivations are collected in the term F_{jNL}^I ,

$$F_{sNL}^I = v_{s-i} m_s \frac{dv_s}{dt} - v_{s-i} m_s r_s \times \frac{d\omega_s}{dt} + \omega_{s-i} m_s r_s \times \frac{dv_s}{dt} \\ + \omega_{s-i} I \frac{d\omega_s}{dt} + m_s v_{s-i} \times \omega_s v_{s-i} + v_{s-i} \omega_s \times \omega_s \times m_s r_s \\ - \omega_s \times m_s r_s \times \omega_s v_{s-i} + \omega_{s-i} \omega_s \times I \omega_s \quad (34)$$

$$\begin{aligned} \mathbf{F}_{\text{INL}}^I = & \mathbf{v}_{fi} m \frac{d\mathbf{v}_f}{dt} - \mathbf{v}_{fi} m \mathbf{r}_f \times \frac{d\boldsymbol{\omega}_f}{dt} + \boldsymbol{\omega}_{fi} m \mathbf{r}_f \times \frac{d\mathbf{v}_f}{dt} \\ & + \boldsymbol{\omega}_{fi} J \frac{d\boldsymbol{\omega}_f}{dt} + m \mathbf{v}_{fi} \times \boldsymbol{\omega}_f \mathbf{v}_{fi} + \mathbf{v}_{fi} \boldsymbol{\omega}_f \times \boldsymbol{\omega}_f \times m \mathbf{r}_f \\ & - \boldsymbol{\omega}_f \times m \mathbf{r}_f \times \boldsymbol{\omega}_f \mathbf{v}_{fi} + \boldsymbol{\omega}_{fi} \boldsymbol{\omega}_f \times J \boldsymbol{\omega}_f \end{aligned} \quad (35)$$

The generalized active forces $\mathbf{F}_s^A$ and $\mathbf{F}_f^A$ are the external linear and angular forces of the spacecraft and the fuel ball, respectively. The generalized active forces $\mathbf{F}_s^A$ of the satellite are represented by the collision force and torque with the fuel ball $\mathbf{f}_{\text{col}}$, the adhesion force and torque $\mathbf{f}_{\text{ad}}$, and the control and disturbance linear and angular 6×1 forces vectors $\mathbf{f}_c$ and $\mathbf{f}_d$ if considered.

For the fuel ball, the only generalized active forces are the collision forces, and in case the tank walls are not considered hydrophobic, the adhesion forces are also included in the generalized active forces:

$$\mathbf{F}_s^A = \mathbf{f}_c + \mathbf{f}_d - [\mathbf{f}_{\text{col}} \quad \mathbf{r}_{\text{col},T} \times \mathbf{f}_{\text{col}}]^T - [\mathbf{f}_{\text{ad}} \quad \mathbf{r}_{\text{col},T} \times \mathbf{f}_{\text{ad}}]^T \quad (36)$$

$$\mathbf{F}_f^A = [\mathbf{f}_{\text{col}} \quad \mathbf{r}_{\text{col},F} \times \mathbf{f}_{\text{col}}]^T + [\mathbf{f}_{\text{ad}} \quad \mathbf{r}_{\text{col},F} \times \mathbf{f}_{\text{ad}}]^T \quad (37)$$

B. Equations of Motion

Kane's equation is defined as

$$\mathbf{F}_j^I + \mathbf{F}_j^A = \mathbf{0} \quad (38)$$

where the index j refers to each of the spacecraft and the fuel ball model. Based on the Eqs. (32–38), the dynamic and kinematic equations of a six-DOF spacecraft that performs translational and rotational motion, in zero gravity, are then written as follows:

$$\begin{aligned} - \left[\dot{\mathbf{v}}_s (\mathbf{v}_{si} m_s \mathbf{v}_{si} - \mathbf{v}_{si} m_s \mathbf{r}_s \times \boldsymbol{\omega}_{s-i} + \boldsymbol{\omega}_{si} m_s \mathbf{r}_s \times \mathbf{v}_{si} + \boldsymbol{\omega}_{si} I \boldsymbol{\omega}_{s-i}) \right. \\ + \mathbf{v}_{si} m_s \frac{d\mathbf{v}_s}{dt} - \mathbf{v}_{si} m_s \mathbf{r}_s \times \frac{d\boldsymbol{\omega}_s}{dt} + \boldsymbol{\omega}_{si} m_s \mathbf{r}_s \times \frac{d\mathbf{v}_s}{dt} + \boldsymbol{\omega}_{si} I \frac{d\boldsymbol{\omega}_s}{dt} \\ + m_s \mathbf{v}_{si} \times \boldsymbol{\omega}_s \mathbf{v}_{si} + \mathbf{v}_{si} \boldsymbol{\omega}_s \times \boldsymbol{\omega}_s \times m_s \mathbf{r}_s - \boldsymbol{\omega}_s \times m_s \mathbf{r}_s \times \boldsymbol{\omega}_s \mathbf{v}_{si} \\ \left. + \boldsymbol{\omega}_{s-i} \boldsymbol{\omega}_s \times I \boldsymbol{\omega}_s \right] + \mathbf{f}_c + \mathbf{f}_d - [\mathbf{f}_{\text{col}} \quad \mathbf{r}_{\text{col},T} \times \mathbf{f}_{\text{col}}]^T \\ - [\mathbf{f}_{\text{ad}} \quad \mathbf{r}_{\text{col},T} \times \mathbf{f}_{\text{ad}}]^T = \mathbf{0} \end{aligned} \quad (39)$$

$$\begin{aligned} - \left[\dot{\mathbf{v}}_f (\mathbf{v}_{fi} m \mathbf{v}_{fi} - \mathbf{v}_{fi} m \mathbf{r}_f \times \boldsymbol{\omega}_{f-i} + \boldsymbol{\omega}_{fi} m \mathbf{r}_f \times \mathbf{v}_{fi} \right. \\ + \boldsymbol{\omega}_{fi} J \boldsymbol{\omega}_{fi}) \mathbf{v}_{fi} m \frac{d\mathbf{v}_f}{dt} - \mathbf{v}_{fi} m \mathbf{r}_f \times \frac{d\boldsymbol{\omega}_f}{dt} + \boldsymbol{\omega}_{fi} m \mathbf{r}_f \times \frac{d\mathbf{v}_f}{dt} \\ + \boldsymbol{\omega}_{fi} J \frac{d\boldsymbol{\omega}_f}{dt} + m \mathbf{v}_{fi} \times \boldsymbol{\omega}_f \mathbf{v}_{fi} + \mathbf{v}_{fi} \boldsymbol{\omega}_f \times \boldsymbol{\omega}_f \times m \mathbf{r}_f \\ \left. - \boldsymbol{\omega}_f \times m \mathbf{r}_f \times \boldsymbol{\omega}_f \mathbf{v}_{fi} + \boldsymbol{\omega}_{f-i} \boldsymbol{\omega}_f \times J \boldsymbol{\omega}_f \right] + [\mathbf{f}_{\text{col}} \quad \mathbf{r}_{\text{col},F} \times \mathbf{f}_{\text{col}}]^T \\ + [\mathbf{f}_{\text{ad}} \quad \mathbf{r}_{\text{col},F} \times \mathbf{f}_{\text{ad}}]^T = \mathbf{0} \end{aligned} \quad (40)$$

All dynamic couplings are highlighted by the common terms in Eqs. (39) and (40) represented by the collision and adhesion forces which occur only when the two bodies are in contact.

VI. Simulations and Discussion

Numerous simulations have been conducted to observe the dynamics of the proposed model and identify its limitations and the features to be improved or optimized in future work. Comparison with previous works results from literature is almost impossible for the lack of such data because this aspect of floating fuel or *wobbling masses* in zero gravity has not been analytically investigated before [30]. In the present paper, the results of three cases are interpreted and discussed. The elastic fuel ball behavior has been observed in both the first and second cases without consideration of adhesion forces, and then in a third case, the adhesive forces in the proximity of the tank walls are taken in account. The results of the latter are compared with CFD results. Cases 1 and 3 are for steady containers and a displacing fuel ball, and case 2 is for orbiting rigid spacecraft. All motion differential equations are solved based on the integration method Runge Kutta45.

A. Case 1: Steady Spacecraft

In a steady spacecraft, the elastic modeled ball is released in zero gravity with an initial velocity of -0.08 m/s in the z -axis direction. The used parameters are listed in Table 1.

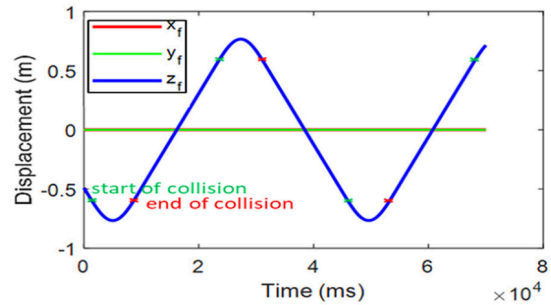


Fig. 8 Liquid elastic ball displacement and collisions start and end marks.

Table 1 Simulation 1 parameters

Model parameters	Values	Model parameters	Values
σ , N/m	0.072	R_T , m	0.8
ρ , kg/m ³	1000	m_s , kg	6362.241
μ , Pa · s	0.001	E_T , GPa	70
m , kg	113.0973	ν_T	0.35
J , kg · m ²	4.071 0 0 0 4.071 0 0 0 4.071	I , kg · m ²	2616.224 0 0 0 3666.224 0 0 0 3666.224
R_f , m	0.3	μ_v	0.1
C_k	8	μ_d	0.5
C_d	5	μ_s	0.7
C_r	0.999	ν_f	0.5
Fuel initial position, m, rad	[0 0 -0.49 0 0 0] ^T	Spacecraft initial position, m, rad	[0 0 0 0 0 0] ^T
Fuel initial velocity, m/s, rad/s	[0 0 -0.08 0 0 0] ^T	Spacecraft initial velocity, m/s, rad/s	[0 0 0 0 0 0] ^T

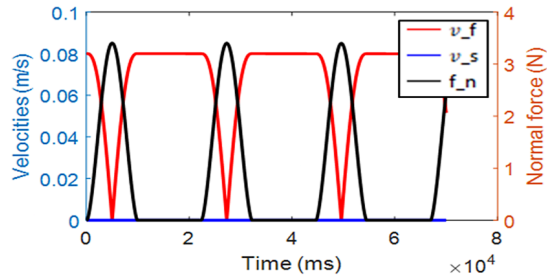


Fig. 9 Collision normal force and velocities magnitudes of the elastic ball and the rigid spacecraft.

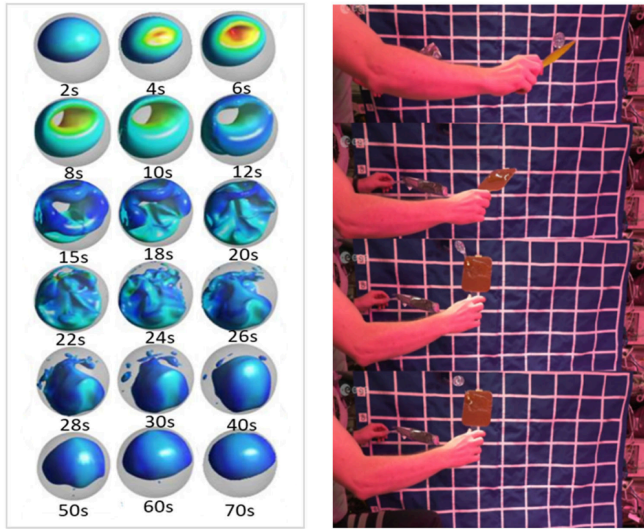


Fig. 10 Coalesced liquid bouncing observations in zero gravity.

When Fig. 8 is read in accordance with Fig. 9, it is seen that the elastic ball reaches the lower tank wall with a constant velocity magnitude equal to 0.08 m/s. Then, at the moment where the surface of the ball is contacting the tank wall as shown by the first

green mark in Fig. 8, the magnitude of the liquid velocity drops gradually from 0.08 to 0 m/s; then, the ball reverses direction, and its velocity increases again to a value very slightly lower than 0.08 m/s. That velocity is then maintained at a constant value during the floating of the elastic ball toward the upper tank wall.

The collision force, on the other hand, at the start of collision starts to increase, opposing the motion of the colliding liquid ball, making it decelerate till stopping. After reaching its maximum, the collision normal force decreases, allowing the fuel ball to regain its original shape and gradually detach from the tank wall till being dominated only by its inertial forces. This cycle keeps repeating; the fuel ball keeps bouncing back and forth along the z axis between the upper and lower tank walls even when tested for lower initial velocities because of the absence of the gravitational forces. However, the cases of higher initial velocities where the liquid bulk breaks after colliding with the tank walls are out of the focus of this paper because there are no available data yet in literature to compare with and the main purpose of this work is to model the behavior of the liquid when it is coalesced as a bulk. We can observe qualitatively the repeated cycle of floating, contacting, deforming, and then detaching from the tank wall as an elastic body in the numerical results in Ref. [31] and experimental results [19] as shown in Figs. 10a and 10b, respectively.

One should not be confused in this case by the fact that the normal force is always at its maximum when the difference between the elastic ball velocity and the spacecraft velocity is null. In fact, the normal force is always at its maximum when the normal component of the difference between the two velocities is 0. In this first case, the velocity has no tangential component because the velocities difference direction is chosen totally coinciding with the collision normal as shown in Fig. 11a.

Arbitrarily changing the initial velocity of the elastic ball from $[0 \ 0 \ -0.08 \ 0 \ 0 \ 0]^T$ to $[-0.047 \ 0.045 \ -0.046 \ 0 \ 0 \ 0]^T$ allows the observation of the elastic ball behavior in a more general case.

The elastic ball trajectory is no more a unidirectional line between the upper and lower tank walls, as can be seen in Figs. 11a and 11d. The contact point P in the arbitrary excitation example is moving in all directions as shown in Figs. 11e and 11f in both the elastic ball frame and the contact frame, contrarily to the unidirectional displacement illustrated in Figs. 11b and 11c.

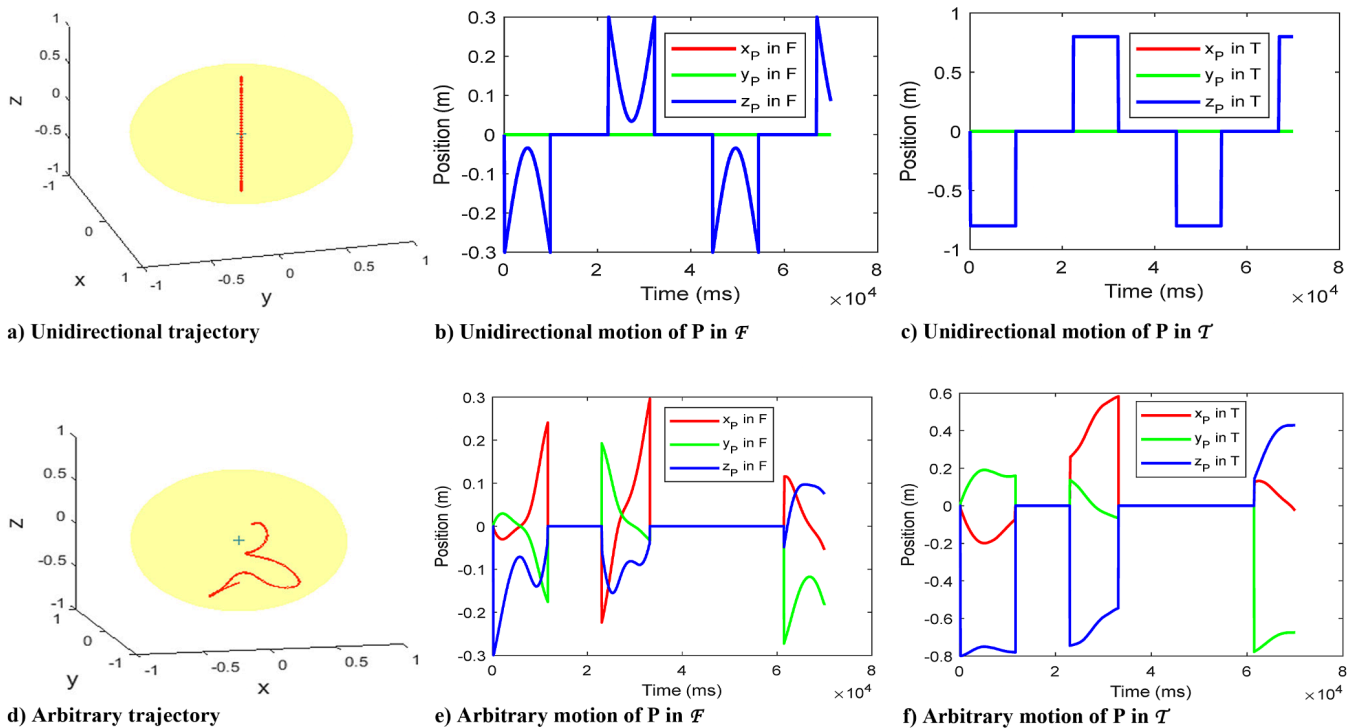


Fig. 11 Liquid trajectory in the tank and P motion in F and T for unidirectional and arbitrary cases.

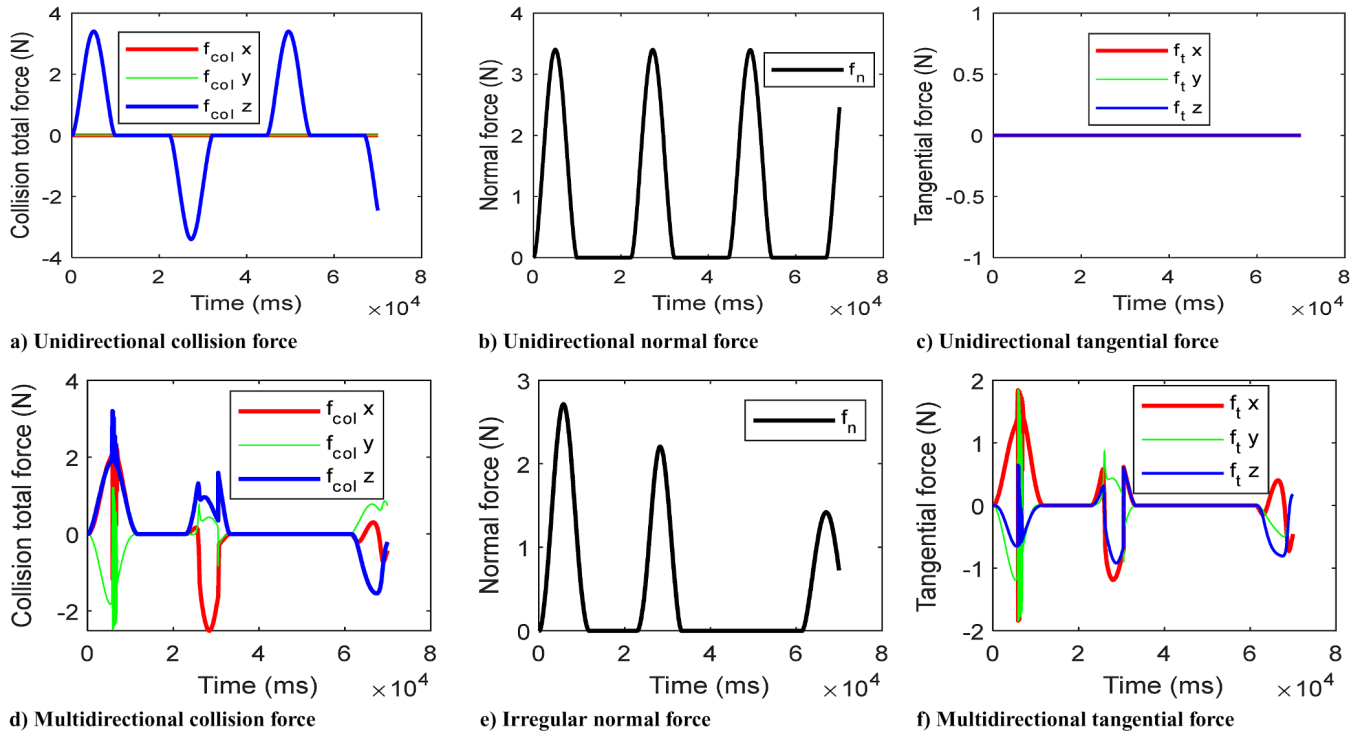


Fig. 12 Collision total, normal and tangential forces for unidirectional and arbitrary cases.

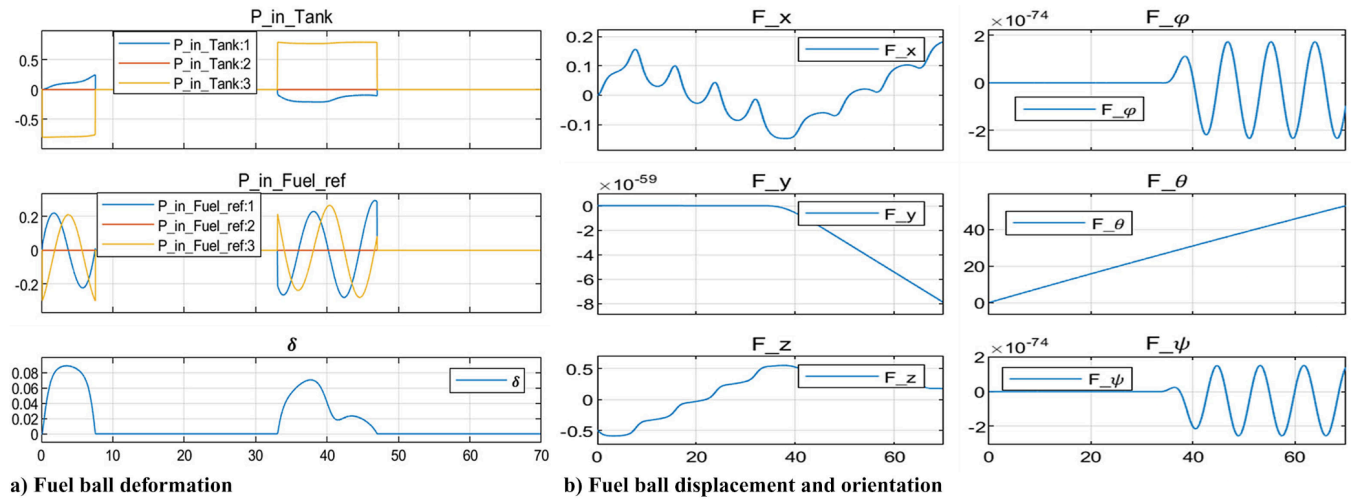


Fig. 13 Fuel ball model response to angular and linear velocities excitation.

Similarly, the collision forces are varying in all directions in the second example Figs 12a and 12d. In Fig. 12f, the tangential component is no more null as it was in Fig. 12c, consequently, the normal component magnitude in Fig. 12e lost its regularity and periodicity compared to the first case in Fig. 12b.

An additional simulation has been done by setting the initial velocity of the elastic ball as $[0 \ 0 \ -0.08 \ 0 \ 0.8 \ 0]^T$ allowed the visualization of the angular displacement of the fuel ball as shown in Fig. 13b.

B. Case 2: Orbiting Spacecraft

In this simulation, Hohmann orbit transfer is to be executed by the spacecraft at rendezvous start from the orbit 1 of radius OR1 to the orbit 2 of radius OR2 as shown in Fig. 14. The escape velocity ΔV at perigee of the transfer trajectory to go from the orbital velocity OV1 to the orbital velocity OV2 is computed in Ref. [32] by the Vis-Viva equation. The used simulation parameters are presented in Table 2. The fuel sphere is assumed to break up every time the embedded deformation length is equal to the initial radius of the fuel ball as

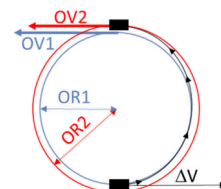


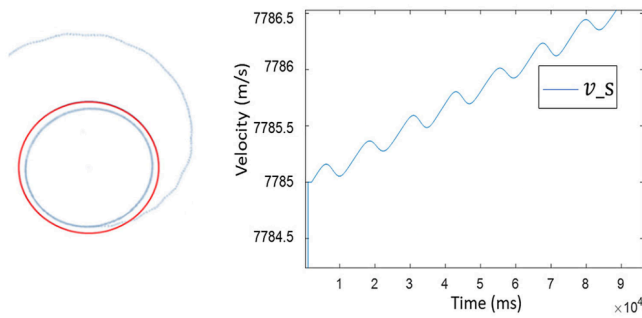
Fig. 14 Hohmann orbit transfer drastic failure.

mentioned in (Sec. II.B). In this simulation, it is assumed that after each breakup the fuel sphere settles at the center of the tank and moves with the spacecraft velocity at the moment of breakup.

Unlike the case of the steady container, the spacecraft six DOF allow us to witness the rigid-flexible coupled dynamics of the fuel ball and the spacecraft and their effect on the spacecraft attitude. The wobbling fuel mass significantly disturbs the spacecraft velocity as shown in Fig. 15b and consequently deorbiting out of the desired trajectory as shown in Fig. 15a. The spacecraft velocity

Table 2 Hohmann orbit transfer simulation parameters

Model parameters	Values	Model parameters	Values
σ , N/m	0.034	R_T , m	0.5
ρ , kg/m ³	874.4	m_s , kg	6362.241
μ , Pa · s	0.001	E_T , GPa	70
m , kg	114.4802105799491	ν_T	0.35
	4.544 0 0		2616.224 0 0
J , kg · m ²	0 4.544 0	I , kg · m ²	0 3666.224 0
	0 0 4.544		0 0 3666.224
R_f , m	0.315	μ_v	0.1
C_k	8	μ_d	0.5
C_d	5	μ_s	0.7
C_r	0.999	v_f	0.5
OV1, m/s	7784	OV2, m/s	7726
OR1, m	6578,140	OR2, m	6676,140
ΔV , m/s	58.5	Fuel initial position, m, rad	[6.578e + 06; 0.05; 0; 0; 0; 0]

**a) Spacecraft trajectory b) Spacecraft velocity disturbance****Fig. 15** Elastic ball wobbling effect on the spacecraft attitude.

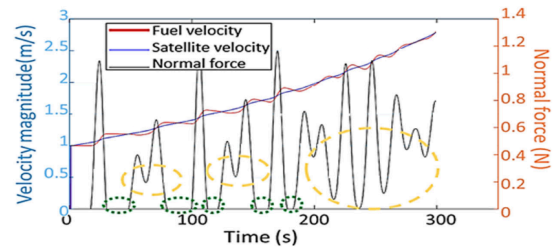
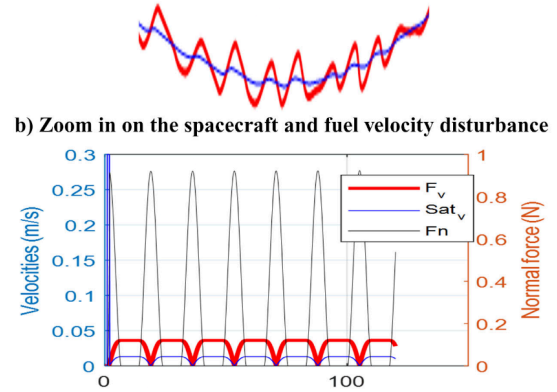
kept oscillating, which, if the spacecraft is equipped with more flexible systems, would undoubtedly lead to partially or totally losing the control of the spacecraft.

Observing the results of numerous simulations of orbiting spacecrafts, it is noticed that, unlike results in the literature, the presented wobbling elastic ball affects the spacecraft attitude by disturbing it sporadically rather than maintaining a continuous contact. As shown as in the example in Fig 16a, the case of an orbiting spacecraft undergoing a ramp augmenting y velocity, the dashed yellow marks show the continuous contact between the fuel ball and the tank wall, and the green ones show the absence of the collision force when the fuel ball is floating inside the tank with no contact. However, these interrupted and unpredictable disturbances caused by the wobbling mass(es) is worth investigation and must not be neglected.

The liquid-rigid coupled dynamics can be seen in Fig. 16b for the ramp response and Fig. 16c for the y -velocity impulsive response, where the collisions between the fuel model and the tank walls are relatively disturbing the spacecraft attitude. In the latter two simulations, the ramp and the impulse values of the spacecraft y -velocity excitation are chosen low to keep the model applicable the long of the simulation far from the range of surface tension breakup as seen in the Hoffman transfer simulation.

C. Case 3: Steady Spacecraft with Adhesive Forces Consideration

This simulation is conducted using the software FLSCON version 11.1.1.3 with additional support from FLOW-3D and HYDR3D version 11.1.1.3. The model parameters are same as those in Table 1, except the initial position of the liquid ball here is $[0 \ 0 \ 0 \ 0 \ 0]^T$ and the tank walls are not considered hydrophobic. The grid resolution used in FLOW-3D simulations is 100000 cells with refinement near the liquid-tank interface to ensure accurate collision and deformation modeling. The simulation run time for the presented cases was 1 h 35 min 45 s on a high-performance workstation (eight-core

**a) Spacecraft and fuel ball velocities and normal force in response to ramp excitation****c) Spacecraft and fuel ball velocities and normal force in response to impulsive excitation****Fig. 16** Spacecraft and fuel ball coupled dynamics.

processor, 64 GB RAM). Grid convergence tests were conducted to confirm that the results are independent of grid resolution.

The results obtained by the proposed novel simplified model and those obtained by the CFD are significantly in agreement as shown in Figs. 17a and 17b. The fuel ball motion and force in zero gravity in a hydrophobic container are proportionally imitated by the proposed model with no signal noise, contrarily to the CFD unduly noised signal. The model still can be enhanced in order to totally eliminate the error; however, the obtained result is promising for such an innovative simplified and manageable mechanical model.

The adhesion effect is obvious where the elastic ball in Fig. 17a is gradually settling down instead of detaching to float away to the opposite tank wall. The magnitude of the adhesion force is not significant compared to the collision force Fig. 18; however, it is enough to attach the liquid ball to the tank wall and prevent it from bouncing away to the opposite wall. The dynamics in such conditions are worth investigation because a big mass of fuel is able to drastically deviate the gravity mass of a spacecraft by oscillating while in unison with the tank wall.